we have $x \overset{iid}{\sim} N(m, \Sigma)$ $\{x_i\}_{i=1}^{n}$ $x_i \in \mathbb{R}^d$

$$L(m \mid x_1, \dots, x_n) = \underbrace{\prod_{i=1}^{n} (2\pi)^{-\frac{d}{2}} |\Sigma|^{-\frac{1}{2}}}_{C} \exp\left(-\frac{1}{2}(x_i - m)^T \Sigma^{-1} (x_i - m)\right)$$

$$\log L(m \mid x_1, \dots, x_n) = \log C + \left(-\frac{1}{2}\right) \sum_{i=1}^{n} (x_i - m)^T \Sigma^{-1} (x_i - m)$$

$$\hat{m}_{ML} = \underset{m}{\operatorname{argmax}} \log L(m \mid x_1, \dots, x_n) = \underset{m}{\operatorname{argmax}} -\sum_{i=1}^{n} \left\| \Sigma^{-\frac{1}{2}} (x_i - m) \right\|_2^2$$

$$\frac{\partial L}{\partial m} = 0 \Rightarrow \sum_{i=1}^{n} 2 \Sigma^{-1} (x_i - m) = 0 \qquad \boxed{\hat{m}_{ML} = \frac{1}{n} \sum_{i=1}^{n} x_i}$$

Similar to Slides.

$$L(\Sigma \mid x_1, \dots, x_n) = \prod_{i=1}^{n} (2\pi)^{-\frac{d}{2}} |\Sigma|^{-\frac{1}{2}} \exp\left(-\frac{1}{2}(x_i - m)^T \Sigma^{-1} (x_i - m)\right)$$

$$\underset{\Sigma}{\operatorname{argmax}} \log L(\Sigma \mid x_1, \dots, x_n) = \underset{\Sigma}{\operatorname{argmax}} -\frac{n}{2} \log|\Sigma| - \frac{1}{2} \sum_{i=1}^{n} \left((x_i - m)^T \Sigma^{-1} (x_i - m)\right)$$

$$= \underset{\Sigma}{\operatorname{argmax}} -\frac{n}{2} \log|\Sigma| - \frac{1}{2} \operatorname{tr}\left(\Sigma^{-1} S(m)\right)$$

using Prop of trace $\operatorname{tr}(ABC) = \operatorname{tr}(BCA)$.

where we define $S(m) = \sum_{i=1}^{n} (x_i - m)(x_i - m)^T$

note Σ is symmetric.

taking derivative

$$\frac{\partial \log|\Sigma|}{\partial \Sigma} = (\Sigma^{-1})^T$$ from matrix cook (uploaded on Canvas)

$$\frac{\partial \, tr(\Sigma^{-1} S)}{\partial \Sigma} = -\Sigma^{-T} S \Sigma^{-T}$$

So combine

$$\frac{\partial L}{\partial \Sigma} = \frac{-n}{2} \Sigma^{-T} + \frac{1}{2} \Sigma^{-T} S \Sigma^{-T} = 0$$

multiply by $2\Sigma^T$ $\longrightarrow$

$$-nI + S\Sigma^{-T} = 0 \quad \Rightarrow \quad \Sigma = \frac{1}{n} S$$

where we est m by $\hat{m}_{ML}$.

Hence,

$$\boxed{\hat{\Sigma}_{ML} = \frac{1}{n} \sum_{i=1}^{n} (x_i - \hat{m}_{ML})(x_i - \hat{m}_{ML})^T}$$

So you can see how both ML est conv to the true m, Σ as n increases.

you can plot the err of $\| \hat{m}_{ML} - m \|^2$ as a example. or mathematically / analytically

you can see that $\hat{m}_{ML} = \frac{1}{n} \sum_{i=1}^{n} x_i$

hence $\hat{m}_{ML} \sim N\left(m, \frac{\Sigma}{n} \right)$ $\Rightarrow$ the err rate in term of how std goes to zero is $O(\frac{1}{\sqrt{n}})$

this means on the order of.

$$\Rightarrow \quad \| \hat{m}_{ML} - m \|_2 = O\left(\frac{1}{\sqrt{n}}\right)$$

for Σ, we have similar rates $\left(\| \hat{\Sigma}_{ML} - \Sigma \|_F^2\right)^{\frac{1}{2}} = O\left(\frac{1}{\sqrt{n}}\right)$

but, no need to derive. we asked to plot errors. when $n \gg$ dim of data.

See <u>notebook.</u>

(d) Let $x \in \mathbb{R}^d$.

$$P(x \mid y = c_0) = (2\pi)^{-\frac{d}{2}} |\Sigma_0|^{-\frac{1}{2}} \exp\left(-\frac{1}{2}(x - m_0)^T \Sigma_0^{-1} (x - m_0)\right)$$

$$P(x \mid y = c_1) = (2\pi)^{-\frac{d}{2}} |\Sigma_1|^{-\frac{1}{2}} \exp\left(-\frac{1}{2}(x - m_1)^T \Sigma_1^{-1} (x - m_1)\right)$$

follow similar steps at Slide 33 in Lecture Prob ML.

$$P(y = c_1 \mid x) \geq P(y = c_0 \mid x)$$

$P(y = c_1) = \pi_1$

$P(y = c_0) = \pi_0$

$$\log P(x \mid y = c_1) - \log P(x \mid y = c_0) + \log \frac{\pi_1}{\pi_0} \geq 0$$

$$-\frac{1}{2}\log|\Sigma_1| + \frac{1}{2}\log|\Sigma_0| \underbrace{- \frac{1}{2}(x - m_1)^T \Sigma_1^{-1} (x - m_1) + \frac{1}{2}(x - m_0)^T \Sigma_0^{-1} (x - m_0)}$$

$$+ \log \frac{\pi_1}{\pi_0} \geq 0$$

denote $h = -\frac{1}{2}\log|\Sigma_1| + \frac{1}{2}\log|\Sigma_0| + \log\frac{\pi_1}{\pi_0} + h_0$

constants

the rest of constants from

$$\underbrace{\frac{1}{2} x^T (\Sigma_0^{-1} - \Sigma_1^{-1}) x}_{\text{quadratic term}} + \underbrace{(m_1^T \Sigma_1^{-1} - m_0 \Sigma_0^{-1}) x}_{\text{first order linear}} + \underbrace{h}_{\text{constant}} \geq 0$$

- Now for same mean, diff cov, we get:

a quadratic term or depending on exact cov quadratic of circle form.

- when cov are the same, the x^2 term disappears.